

DATABASE MANAGEMENT SYSTEMS (IT 2040)

Question 1

What is a view in a database?

- A) A physical table stored permanently on disk
- B) A virtual table derived from base tables that doesn't necessarily exist in physical form
- C) A backup copy of a table
- D) An index on a table

Answer: B

Question 2

Which statement about views is FALSE?

- A) Views can simplify complex queries
- B) Views can provide security by restricting data access
- C) All views are updateable
- D) Views don't physically store data

Answer: C.

Question 3

Given a view created as:

```
CREATE VIEW dept_summary AS
SELECT deptId, COUNT(*) as empCount, AVG(salary) as avgSalary
FROM Employee
GROUP BY deptId;
```

Can you execute: UPDATE dept_summary SET avgSalary = 50000 WHERE deptId = 10;?

- A) Yes, it updates the average salary
- B) No, views with aggregate functions are not updateable
- C) Yes, but it updates all employee salaries
- D) Yes, but only if there's an INSTEAD OF trigger

Answer: B (or D if trigger exists)

Question 4

What happens when you drop a base table that a view depends on?

- A) The view is automatically dropped
- B) The view remains but queries fail
- C) The view is converted to a regular table
- D) Depends on the DBMS

Answer: D

Question 5

In T-SQL, which is the correct way to declare a variable?

- A) VAR @empId INT;
- B) DECLARE @empId INT;
- C) SET @empId INT;
- D) INT @empId;

Answer: B

Question 6

What is the initial value of a declared T-SQL variable?

- A) 0 for numbers, empty string for text
- B) NULL
- C) Undefined
- D) Must be initialized immediately

Answer: B

Question 7

How do you assign a value to a variable in T-SQL?

- A) @empId = 100;
- B) SET @empId = 100;
- C) LET @empId = 100;
- D) ASSIGN @empId = 100;

Answer: B

Question 8

What does the BREAK statement do in a WHILE loop?

- A) Skips current iteration and continues to next
- B) Exits the innermost WHILE loop completely
- C) Pauses execution temporarily
- D) Exits all nested loops

Answer: B

Question 9

What does the CONTINUE statement do in a WHILE loop?

- A) Exits the loop
- B) Skips remaining statements in current iteration and restarts the loop
- C) Continues to the next loop
- D) Does nothing

Answer: B

Question 10

What is the key difference between a stored function and a stored procedure?

- A) Functions can have parameters, procedures cannot
- B) Functions must return a single value, procedures may return multiple values via OUTPUT
- C) Procedures execute faster
- D) No significant difference

Answer: B

Question 11

Can a stored function use OUTPUT parameters?

- A) Yes, just like procedures
- B) No, functions only use IN parameters and RETURN value
- C) Only for table-valued functions

- D) Yes, but only one OUTPUT parameter

Answer: B

Question 12

How do you call a stored function in T-SQL?

- A) EXEC functionName(parameters);
- B) CALL functionName(parameters);
- C) SELECT dbo.functionName(parameters);
- D) EXECUTE FUNCTION functionName(parameters);

Answer: C

Question 13 Given:

```
CREATE FUNCTION dbo.getBonus(@salary FLOAT) RETURNS FLOAT
```

```
AS
```

```
BEGIN
```

```
    RETURN @salary * 0.10
```

```
END
```

Which call is INCORRECT?

- A) SELECT dbo.getBonus(50000);
- B) SELECT dbo.getBonus(salary) FROM Employee;
- C) EXEC dbo.getBonus 50000;
- D) SELECT *, dbo.getBonus(salary) as bonus FROM Employee;

Answer: C

Question 14

What is the default parameter mode for stored procedures if not specified?

- A) OUTPUT
- B) IN
- C) INOUT
- D) Must always be specified

Answer: B

Question 15

Given a procedure:

```
CREATE PROCEDURE calcStats(@deptId INT, @maxSal FLOAT OUTPUT, @minSal  
FLOAT OUTPUT)
```

```
AS
```

```
BEGIN
```

```
    SELECT @maxSal=MAX(salary), @minSal=MIN(salary)
```

```
    FROM Employee WHERE deptId=@deptId
```

```
END
```

Which call is CORRECT?

- A) EXEC calcStats 10, @max, @min;
- B) DECLARE @max FLOAT, @min FLOAT; EXEC calcStats 10, @max OUTPUT, @min OUTPUT;
- C) SELECT calcStats(10, @max, @min);
- D) CALL calcStats(10, @max, @min);

Answer: B

Question 16

What is a trigger in a database?

- A) A scheduled task that runs periodically
- B) A special stored procedure that automatically executes in response to DML/DDL events
- C) A constraint on a table
- D) A backup mechanism

Answer: B

Question 17

What are the INSERTED and DELETED virtual tables in triggers?

- A) Permanent system tables
- B) Temporary tables created during trigger execution storing affected rows
- C) Backup tables
- D) User-created tables

Answer: B

Question 18

For an INSERT operation, what data appears in trigger tables?

- A) INSERTED has new rows, DELETED is empty
- B) DELETED has new rows, INSERTED is empty
- C) Both tables have the new rows
- D) Both tables are empty

Answer: A

Question 19

For an UPDATE operation, what appears in INSERTED and DELETED tables?

- A) INSERTED has old values, DELETED has new values
- B) INSERTED has new values, DELETED has old values
- C) Both have new values
- D) Only INSERTED has values

Answer: B

Question 20

For a DELETE operation, what appears in trigger tables?

- A) INSERTED has deleted rows, DELETED is empty
- B) DELETED has deleted rows, INSERTED is empty
- C) Both tables have deleted rows
- D) Both tables are empty

Answer: B

Question 21

What is the difference between AFTER and INSTEAD OF triggers?

- A) No difference, just different syntax
- B) AFTER executes after DML completes, INSTEAD OF executes in place of DML
- C) AFTER is faster
- D) INSTEAD OF is only for DDL

Answer: B

Question 22

Can AFTER triggers be defined on views?

- A) Yes, on any view
- B) No, AFTER triggers cannot be defined on views
- C) Only on updateable views
- D) Only with special permissions

Answer: B

Question 23

How many INSTEAD OF triggers can be defined per INSERT operation on a table?

- A) Unlimited
- B) At most one
- C) Exactly two (before and after)
- D) None - INSTEAD OF only for views

Answer: B

Question 24

Given a trigger:

```
CREATE TRIGGER prevent_delete
```

```
ON Employee
```

```
INSTEAD OF DELETE
```

```
AS
```

```
BEGIN
```

```
    PRINT 'Deletion not allowed'
```

```
END
```

What happens when you execute DELETE FROM Employee WHERE eid = 100;?

- A) Employee 100 is deleted
- B) Message printed, but employee is NOT deleted
- C) Error occurs
- D) All employees are deleted

Answer: B

Question 25

What does ROLLBACK TRANSACTION do in a trigger?

- A) Saves changes permanently
- B) Undoes the entire transaction including the triggering operation
- C) Only undoes trigger code
- D) Creates a backup

Answer: B

Question 26

Which trigger is better for implementing validation logic?

- A) AFTER trigger with ROLLBACK if validation fails
- B) INSTEAD OF trigger that only executes DML if validation passes
- C) Both are equally good
- D) Neither - use CHECK constraints

Answer: B

Question 27

Can triggers call other stored procedures?

- A) No, triggers are isolated
- B) Yes, triggers can execute procedures
- C) Only AFTER triggers can
- D) Only with special permissions

Answer: B

Question 28

What happens if a trigger raises an error?

- A) Error is ignored, operation continues
- B) Trigger code stops, but DML operation completes
- C) Entire transaction is rolled back (for AFTER triggers)
- D) Only the trigger is rolled back

Answer: C

Question 29

Which is a valid advantage of using views?

- A) Faster query execution
- B) Security - restrict user access to specific columns/rows
- C) Automatic backup
- D) Increase storage space

Answer: B

Question 30

What is a disadvantage of views?

- A) Too much disk space usage
- B) Cannot query multiple tables
- C) Performance overhead - queries on views must be translated to base table queries
- D) Limited to one view per table

Answer: C

SECTION B: MULTIPLE ANSWER QUESTIONS (Select ALL that apply) - 10 Questions**Question 31**

Which types of views are generally NOT updateable? **(Select all that apply)**

- A) Views with aggregate functions (SUM, AVG, COUNT)
- B) Views joining multiple tables
- C) Views with DISTINCT
- D) Simple SELECT from single table
- E) Views with GROUP BY
- F) Views with subqueries in SELECT

Answers: A, B, C, E, F

Question 32

Which statements about T-SQL variables are TRUE? **(Select all that apply)**

- A) Variable names must start with @
- B) Variables must be declared before use
- C) Variables are initialized to NULL by default
- D) Variables can store query results
- E) Variables persist across sessions
- F) Can use variables in WHERE clauses

Answers: A, B, C, D, F

Question 33

Which control-of-flow statements are available in T-SQL? **(Select all that apply)**

- A) IF...ELSE
- B) WHILE
- C) FOR loop
- D) BREAK
- E) CONTINUE
- F) SWITCH...CASE

Answers: A, B, D, E

Question 34

Advantages of stored procedures include: **(Select all that apply)**

- A) Reduced network traffic (logic executes on server)
- B) Improved performance (pre-compiled execution plan)
- C) Enhanced security (users don't need direct table access)
- D) Code reusability across applications
- E) Automatic data backup
- F) Centralized business logic

Answers: A, B, C, D, F

Question 35

Which are valid characteristics of stored functions? **(Select all that apply)**

- A) Must return a value
- B) Can use OUTPUT parameters
- C) Can be called in SELECT statements
- D) Can modify database tables
- E) Parameters are input-only (IN mode)
- F) Can call other functions

Answers: A, C, E, F

Question 36

What can trigger a DML trigger? **(Select all that apply)**

- A) INSERT statement
- B) UPDATE statement ☒ ☐
- C) DELETE statement ☒
- D) SELECT statement
- E) TRUNCATE statement
- F) MERGE statement

Answers: A, B, C, F

Question 37

For a trigger on Employee table with UPDATE operation, INSERTED and DELETED tables contain: **(Select all that apply)**

- A) INSERTED: new values after update
- B) DELETED: old values before update
- C) Both tables have same structure as Employee table
- D) INSERTED: only changed columns
- E) DELETED: only for deleted rows
- F) Both tables can have multiple rows (if multiple rows updated)

Answers: A, B, C, F

Question 38

When should you use INSTEAD OF triggers? **(Select all that apply)**

- A) To update views that join multiple tables
- B) To implement custom INSERT logic
- C) To prevent certain operations entirely
- D) To audit changes after they occur
- E) To override default DML behavior
- F) When AFTER triggers are not sufficient

Answers: A, B, C, E, F

Question 39

Which scenarios require stored procedures over functions? **(Select all that apply)**

- A) Need to return multiple values
- B) Need to modify database tables
- C) Need to use OUTPUT parameters
- D) Need to call from SELECT statement
- E) Need to handle transactions
- F) Need to execute dynamic SQL

Answers: A, B, C, E, F

Question 40

Best practices for triggers include: **(Select all that apply)**

- A) Keep trigger logic simple and fast
- B) Avoid lengthy transactions in triggers
- C) Use triggers for complex business logic
- D) Document what triggers do
- E) Consider using procedures called from triggers for complex logic 
- F) Avoid recursive triggers

Answers: A, B, D, E, F

SECTION C: ADVANCED SCENARIO-BASED QUESTIONS - 10 Questions**Question 41**

Scenario: Banking system with Account(accountNo, balance), Transaction(transId, accountNo, amount, type, date).

Requirement: Automatically update account balance when transactions are inserted.

Which trigger is MOST robust?

-- Option A

CREATE TRIGGER update_balance

ON Transaction

AFTER INSERT

AS

```
BEGIN
    UPDATE Account
    SET balance = balance +
        (SELECT SUM(CASE WHEN type='credit' THEN amount ELSE -amount END)
         FROM inserted WHERE accountNo = Account.accountNo)
    WHERE accountNo IN (SELECT accountNo FROM inserted)
END
```

-- Option B

```
CREATE TRIGGER update_balance
ON Transaction
AFTER INSERT
AS
BEGIN
    UPDATE a
    SET a.balance = a.balance +
        CASE WHEN i.type='credit' THEN i.amount ELSE -i.amount END
    FROM Account a
    INNER JOIN inserted i ON a.accountNo = i.accountNo
END
```

Answer: B

Question 42

Scenario: Employee(eid, name, salary, managerId) with self-referencing managerId.

Requirement: Ensure employee salary never exceeds manager's salary.

Which trigger handles UPDATE correctly?

-- Option A

```
CREATE TRIGGER check_salary
ON Employee
AFTER UPDATE
AS
BEGIN
```

```

IF EXISTS (
    SELECT * FROM inserted i
    JOIN Employee m ON i.managerId = m.eid
    WHERE i.salary > m.salary
)
BEGIN
    ROLLBACK TRANSACTION
    RAISERROR('Salary exceeds manager', 16, 1)
END
END

-- Option B
CREATE TRIGGER check_salary
ON Employee
AFTER UPDATE
AS
BEGIN
    IF UPDATE(salary) OR UPDATE(managerId)
    BEGIN
        IF EXISTS (
            SELECT * FROM inserted i
            JOIN Employee m ON i.managerId = m.eid
            WHERE i.salary > m.salary AND i.managerId IS NOT NULL
        )
        BEGIN
            ROLLBACK TRANSACTION
            RAISERROR('Salary exceeds manager', 16, 1)
        END
    END
END
END

```

Answer: B

Question 43

Scenario: Library: Book(isbn, title), Copy(copyId, isbn), Loan(loanId, copyId, memberId, loanDate, returnDate).

Create view showing books currently on loan:

```
CREATE VIEW books_on_loan AS
SELECT b.isbn, b.title, c.copyId, l.memberId, l.loanDate
FROM Book b
JOIN Copy c ON b.isbn = c.isbn
JOIN Loan l ON c.copyId = l.copyId
WHERE l.returnDate IS NULL;
```

Can you execute: UPDATE books_on_loan SET title = 'New Title' WHERE isbn = '12345';?

- A) Yes, updates the book title
- B) No, view joins multiple tables
- C) Yes, with INSTEAD OF trigger
- D) Both B and C

Answer: D

Question 44

Scenario: Create function to calculate employee's total work percentage across all departments.

Given: Employee(eid, name), Works(eid, deptId, pct_time) **Which function is correct?**

-- Option A

```
CREATE FUNCTION dbo.getTotalPct(@emplId INT) RETURNS FLOAT
AS
BEGIN
    DECLARE @total FLOAT
    SELECT @total = SUM(pct_time)
    FROM Works
    WHERE eid = @emplId
    RETURN @total
END
```


-- Option B

```
CREATE FUNCTION dbo.getTotalPct(@empld INT) RETURNS FLOAT
AS
BEGIN
    DECLARE @total FLOAT
    SELECT @total = ISNULL(SUM(pct_time), 0)
    FROM Works
    WHERE eid = @empld
    RETURN @total
END
```

Answer: B

Question 45

Scenario: Product(pid, name, price, stock), Sale(saleId, pid, quantity, saleDate).

Requirement: Prevent sales when stock insufficient, and reduce stock when sale succeeds.

Best trigger design:

```
CREATE TRIGGER process_sale
ON Sale
INSTEAD OF INSERT
AS
BEGIN
    -- Check stock sufficient
    IF EXISTS (
        SELECT * FROM inserted i
        JOIN Product p ON i.pid = p.pid
        WHERE i.quantity > p.stock
    )
    BEGIN
        RAISERROR('Insufficient stock', 16, 1)
        RETURN
    END
```

-- Insert sale

INSERT INTO Sale SELECT * FROM inserted

-- Reduce stock

UPDATE p

SET p.stock = p.stock - i.quantity

FROM Product p

JOIN inserted i ON p.pid = i.pid

END

Why use INSTEAD OF instead of AFTER?

- A) AFTER cannot access INSERTED table
- B) INSTEAD OF allows checking before operation and combining logic
- C) AFTER is slower
- D) INSTEAD OF is required for stock updates

Answer: B

Question 46

Scenario: HR system - Employee(eid, name, salary, deptId), Department(deptId, name, budget).

Create procedure to give raise to all employees in a department, ensuring total salary doesn't exceed budget:

CREATE PROCEDURE giveRaise(

 @deptId INT,

 @raisePercent FLOAT,

 @success BIT OUTPUT

)

AS

BEGIN

 DECLARE @totalSalary FLOAT, @budget FLOAT, @newTotal FLOAT

 SELECT @totalSalary = SUM(salary)

```
FROM Employee WHERE deptId = @deptId
```

```
SELECT @budget = budget FROM Department WHERE deptId = @deptId
```

```
SET @newTotal = @totalSalary * (1 + @raisePercent/100)
```

```
IF @newTotal <= @budget
```

```
BEGIN
```

```
    UPDATE Employee
```

```
    SET salary = salary * (1 + @raisePercent/100)
```

```
    WHERE deptId = @deptId
```

```
    SET @success = 1
```

```
END
```

```
ELSE
```

```
    SET @success = 0
```

```
END
```

To call this procedure:

```
DECLARE @result BIT
```

```
EXEC giveRaise 10, 5.0, @result OUTPUT
```

```
IF @result = 1
```

```
    PRINT 'Raise applied'
```

```
ELSE
```

```
    PRINT 'Raise would exceed budget' This pattern
```

demonstrates:

- A) Transaction management
- B) Business rule enforcement in stored procedures
- C) OUTPUT parameter usage
- D) All of the above

Answer: D

Question 47

Scenario: Audit system tracks all changes to sensitive Employee salary field.

Tables: Employee(eid, name, salary), SalaryAudit(auditId IDENTITY, eid, oldSalary, newSalary, changeDate, changedBy)

Create comprehensive audit trigger:

```
CREATE TRIGGER audit_salary
ON Employee
AFTER UPDATE
AS
BEGIN
    IF UPDATE(salary)
    BEGIN
        INSERT INTO SalaryAudit (eid, oldSalary, newSalary, changeDate, changedBy)
        SELECT
            i.eid,
            d.salary,
            i.salary,
            GETDATE(),
            SYSTEM_USER
        FROM inserted i
        JOIN deleted d ON i.eid = d.eid
        WHERE i.salary <> d.salary
    END
END
```

What does UPDATE(salary) check?

- A) If salary is NULL
- B) If salary column was included in UPDATE statement (even if value unchanged)
- C) If salary value actually changed
- D) If salary is valid

Answer: B

Question 48

Scenario: Student registration system prevents overbooking courses.

Tables: Course(courseId, title, maxStudents), Enrollment(studentId, courseId, enrollDate)

Create trigger to enforce maximum:

```
CREATE TRIGGER prevent_overbook
ON Enrollment
INSTEAD OF INSERT
AS
BEGIN
    DECLARE @courseId INT, @maxStudents INT, @currentCount INT

    SELECT @courseId = courseId FROM inserted
    SELECT @maxStudents = maxStudents FROM Course WHERE courseId = @courseId
    SELECT @currentCount = COUNT(*) FROM Enrollment WHERE courseId = @courseId

    IF @currentCount < @maxStudents
        INSERT INTO Enrollment SELECT * FROM inserted
    ELSE
        RAISERROR('Course is full', 16, 1)
END
```

What's the flaw in this trigger?

- A) Should use AFTER instead of INSTEAD OF
- B) Doesn't handle multiple students enrolling simultaneously
- C) Should check maxStudents before counting
- D) Missing transaction handling

Answer: B

Question 49

Scenario: Complex view joins Student, Enrollment, and Course tables. User tries to update course name through the view.

What happens?

```
CREATE VIEW student_courses AS
SELECT s.sid, s.name AS studentName, c.courseld, c.name AS courseName
FROM Student s
JOIN Enrollment e ON s.sid = e.sid
JOIN Course c ON e.courseld = c.courseld;
```

-- User executes:

```
UPDATE student_courses SET courseName = 'Database Systems' WHERE courseld =
'CS101'; Result:
```

- A) Update succeeds and course name changes
- B) Update fails - views with joins are not directly updateable
- C) Update succeeds but only for one student's enrollment
- D) Update succeeds with INSTEAD OF trigger

Answer: B

Question 50

Scenario: HR database: Employee(eid, name, salary, deptId), Department(deptId, deptName, budget).

Requirement: Create a function to calculate what percentage of department budget is spent on salaries.

Correct function:

```
CREATE FUNCTION dbo.getSalaryPercentage(@deptId INT)
RETURNS FLOAT
AS
BEGIN
```

```
DECLARE @totalSalary FLOAT, @budget FLOAT, @percentage FLOAT
```

```
SELECT @totalSalary = SUM(salary)
```

```
FROM Employee
```

```
WHERE deptId = @deptId
```

```
SELECT @budget = budget
```

```
FROM Department
```

```
WHERE deptId = @deptId
```

```
IF @budget > 0
```

```
    SET @percentage = (@totalSalary / @budget) * 100
```

```
ELSE
```

```
    SET @percentage = 0
```

```
RETURN @percentage
```

```
END
```

This function is called using:

- A) EXEC dbo.getSalaryPercentage 101
- B) SELECT dbo.getSalaryPercentage(101)
- C) CALL getSalaryPercentage(101)
- D) EXECUTE FUNCTION getSalaryPercentage(101)

Answer: B

Question 51

Which of the following statements are true related to constraints? (Select one or more)

- A) If a subtype participates in a relationship that is the same as the other subtypes, that relationship could be added to the supertype.
- B) When an entity belongs to only one subtype in the hierarchy, the relationship is total and disjoint.
- C) A bowler and batsman (wicketkeeper is also a batsman) which are subtypes of a cricketer type is total and disjoint.

- D) When an entity instance may be a member of multiple subtypes, or does not have to be a member of a subtype, the specialization is overlapping and total.

Answer: A

Question 52

Consider the following table definition: CREATE

```
TABLE Product ( pid CHAR(4) PRIMARY KEY,  
pname VARCHAR(30), manufactureDate  
DATETIME DEFAULT GETDATE(),  
qty INT,  
CONSTRAINT pid_chk CHECK (pid LIKE 'PSTO-510-5[0-5]-577'),  
CONSTRAINT qty_chk CHECK (qty >= 0)  
);
```

Which of the following statements are true related to the above definition? (Select one or more)

- A) Consider a product with a quantity (qty) of 10. An update to deduct 15 from the available quantity in the above table will not be successful.
- B) "T12" is a valid pid that could be stored in the Product table.
- C) "P333" is a valid pid that could be stored in the Product table.
- D) Executing the following insert statement will cause an error: INSERT INTO Product(pid, qty) VALUES ('T000', 50)
- E) After executing the following insert statement in the above table, the row inserted will have two null values: INSERT INTO Product(pid, qty) VALUES ('T000', 50)

Answer: A, D

Question 53

Consider the following table definition: CREATE

```
TABLE Product ( pid CHAR(4) PRIMARY KEY,  
pname VARCHAR(30), manufactureDate  
DATETIME DEFAULT GETDATE(),
```



```
Qty INT,  
CONSTRAINT pid_chk CHECK (pid LIKE '[P|S|T][0-5][0-5][0-5]'),  
CONSTRAINT qty_chk CHECK (qty > 0)  
);
```

Which of the following statements are true related to the above definition? (Select one or more)

- A) Consider a product with a quantity (qty) of 10. An update written to deduct 15 from the available quantity in the above table will not be successful.
- B) "T12" is a valid pid that could be stored in the Product table.
- C) "P333" is a valid pid that could be stored in the Product table.
- D) Executing the following insert statement will cause an error: INSERT INTO Product(pid, qty) VALUES ('T000', 50)
- E) After executing the following insert statement in the above table, the row inserted will have two null values: INSERT INTO Product(pid, qty) VALUES ('T000', 50)

Answer: A, C

Question 54

```
Consider the following table definition: CREATE  
TABLE Product ( pid CHAR(4) PRIMARY KEY,  
pname VARCHAR(30), manufactureDate  
DATETIME DEFAULT GETDATE(),  
qty INT,  
CONSTRAINT pid_chk CHECK (pid LIKE 'P|S|T[0-5][0-5]10-5|10-577'),  
CONSTRAINT qty_chk CHECK (qty > 0)  
);
```

Which of the following statements are TRUE related to the above definition? (Select one or more)

- A) Consider a product with a quantity (qty) of 10. An update statement written to deduct 15 from the available quantity in the above table will not be successful.
- B) T12 is a valid pid that could be stored in the Product table
- C) P333 is a valid pid that could be stored in the Product table
- D) Executing the following insert statement will cause an error:

Answer: A

Question 55

Which of the following statements are true related to constraints? (Select one or more)

- A) If a subtype participates in a relationship that is the same as the other subtypes, that relationship could be added to the supertype
- B) When an entity belongs to only one subtype in the hierarchy, the relationship is total and disjoint
- C) When an entity instance may be a member of multiple subtypes or it does not have to be a member of a subtype, the specialization is overlapping and total
- D) A bowler and batsman (assume wicket keeper is also a batsman), which are subtypes of a cricketer type, is total and disjoint

Answer: A

Question 56

Which of the following statements are TRUE related to constraints? (Select one or more)

- A) A bowler and batsman (assume wicket keeper is also a batsman), which are subtypes of a cricketer type, is total and disjoint
- B) If a subtype participates in a relationship that is the same as the other subtypes, that relationship could be added to the supertype
- C) When an entity belongs to only one subtype in the hierarchy, the relationship is total and disjoint
- D) When an entity instance may be a member of multiple subtypes or it does not have to be a member of a subtype, the specialization is overlapping and total

Answer: B

Question 57

Consider table definition: CREATE TABLE Product
(pid CHAR(4) PRIMARY KEY, pname
VARCHAR(30), manufactureDate DATETIME
DEFAULT GETDATE(),
Qty INT,
CONSTRAINT pid_chk CHECK (pid LIKE '[P|S|T][0-5][0-5][0-5]'),
CONSTRAINT qty_chk CHECK (qty > 0)
);

Which statements are true? (Select one or more)

- A) T12 is a valid pid that could be stored in the Product table.
- B) Consider a product with a quantity (qty) of 10. Updating the table to deduct 15 will not be successful.
- C) After executing INSERT INTO Product(pid, qty) VALUES('T000', 50), the row inserted will have two null values.
- D) P333 is a valid pid that could be stored in the Product table.
- E) Executing INSERT INTO Product(pid, qty) VALUES('T000', 50) will cause an error

Answer: B, D
